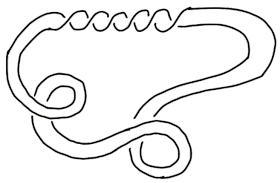
DNA Topology
Introductory Exercises

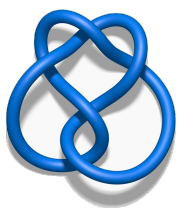
1. Select a knot from the table below. Calculate the writhe of both the knot and its mirror image from the diagram.

2. Calculate $Lk = Tw + Wr$ for the following diagram. Convince yourself the calculation is correct by 'pulling the ribbon tight.'

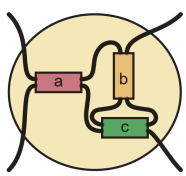


3. Explain why recombination at sites in direct repeat along circular DNA results in a two-component catenane, whereas recombination at sites in inverted repeat produces a single-component knot.

4. Consider the following diagram of the twist knot 5_2. Determine the product of crossing change and both coherent and non-coherent band surgery along the bottom-most crossing in the diagram.



5. Find diagrams for rational tangles of the form $(0), (\infty), (3), (\frac{1}{3})$, and $(\frac{2}{5})$ or your favorite rational number.



$[a, b, c] = a - \frac{1}{b - \frac{1}{c}}$

6. It is a fact from knot theory that any knot obtained by a coherent band surgery on the two-component unlink 0_2 has determinant a square. Use this fact to completely determine which knots K (of crossing number at most eight) satisfy the following tangle equations

$$N(T + \infty) = 0_2$$

$$N(T + 0) = K$$

where T is any tangle (rational or otherwise).

7. Use the Jones polynomial to determine that there is no band surgery relating the knots 4_1 and $11n77$.

Knot theory resources online:

- The Knot Atlas: `https://katlas.org/`
- KnotInfo: `https://knotinfo.math.indiana.edu/`
- Knot Plot: `https://knotplot.com/`

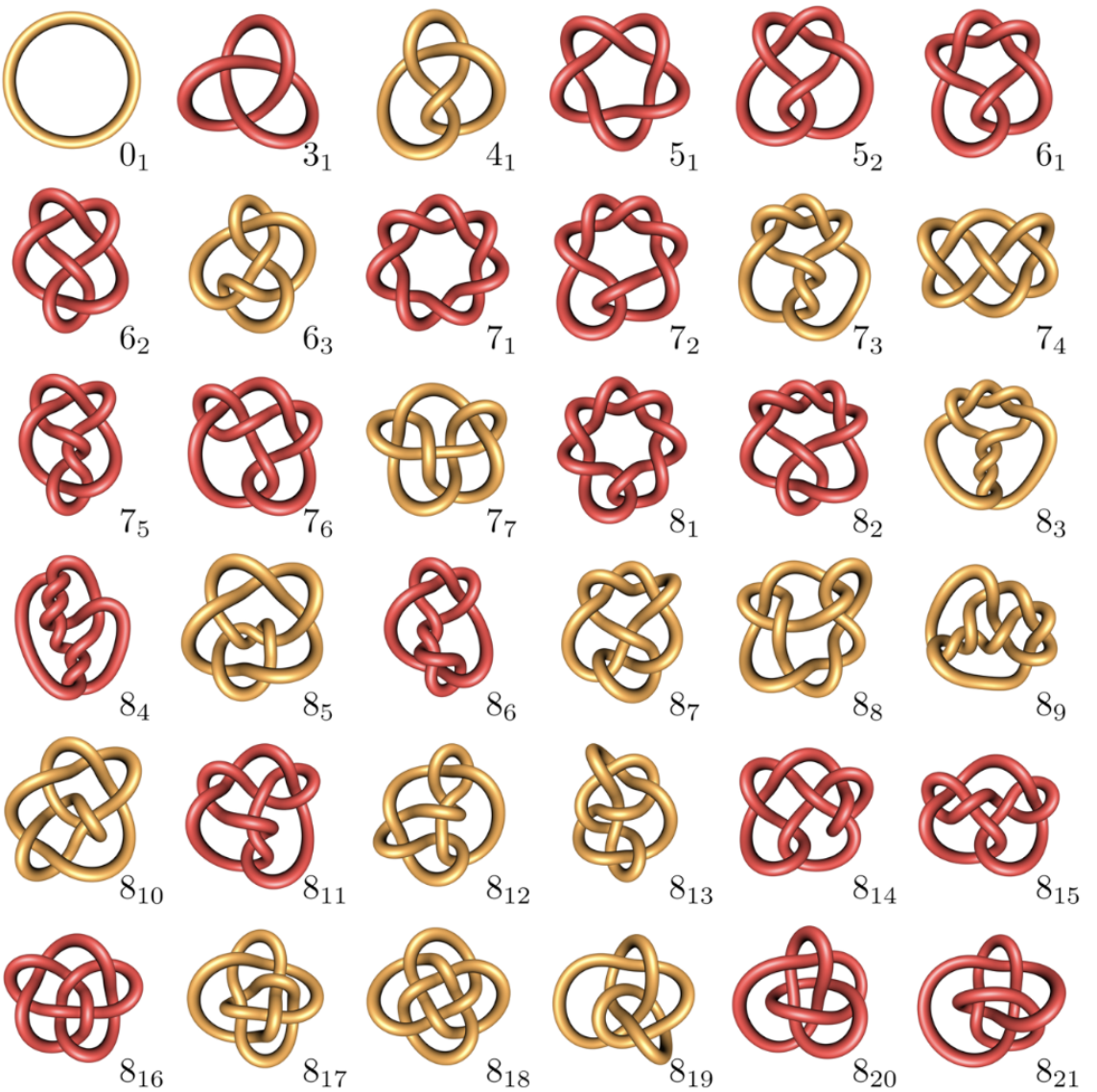


Figure 1: Chiral knot table from Brasher et. al., *Biochem. Soc. Trans.* (2013).